SOFTWARE ENGINEERING MINI-PROJECT

# Software Test Plan

Task Manager Application — STP

| Document control | Value |
|---|---|
| Project | Task Manager Application v1.0 |
| Version | 1.0 |
| Prepared / revised | 15 October 2026 |
| Status | Working baseline — approved |
| Technology baseline | Next.js (React), Node.js/Express REST API, MongoDB/Mongoose |
| Team | Akash Kiran Vernekar (PES1UG24AM338); Vismay H S (PES1UG24AM325); Basawaraj Panchal (PES1UG24AM350); Trisha G Shetty (PES1UG24AM308) |
| Submission deadline | 30 October 2026 |

*Control note: These documents define the proposed scope, architecture and verification plan. They do not claim the application is already implemented, deployed, tested, or approved. Update revision history, actual Jira keys, build status, test evidence and sign-offs as work is completed.*

# Document Control and Revision History

| Version | Date | Owner | Change |
|---|---|---|---|
| 1.0 | 15 October 2026 | Project team | Consolidated test baseline aligned to SRS v1.0 and SAD v1.0; expanded React/Next.js, API, MongoDB, authorization, performance, CI and traceability coverage. |

Approval status: pending. The plan defines tests to run; all test states are Planned until executed and supported by evidence. Performance and monthly availability targets are not reported as passed without measurements.

# 1. Introduction

## 1.1 Purpose

This STP defines the test scope, approach, environment, responsibilities, test cases, entry/exit criteria, risks and reporting for Task Manager Application v1.0. SRS is the requirements source; SAD describes components and sequence flows.

## 1.2 Scope

Test Next.js/React UI, Express REST API, authentication and authorization, Mongoose models/repositories, MongoDB persistence, task/project/board workflows, collaboration and reporting, as well as CI build/test checks. Cloud-provider internals and offline use are out of scope.

# 2. Test Items

- Next.js pages and reusable React components: form validation, board and filters, empty/error/loading states.
- Express routes/controllers/middleware and business services: auth, project/task, collaboration/notification, search/reporting.
- Mongoose schema/repository integration with a dedicated MongoDB test database.
- Browser/API integration, CSV export, authorization boundaries and data integrity.
- GitHub Actions CI jobs and merge readiness checks.

# 3. Features to be Tested and Requirements Coverage

| Requirement IDs | Feature group | Main validation |
|---|---|---|
| TM-F-001..003 | Registration/login/profile | Valid/invalid inputs, duplicate email, hashing, token expiry, persistence. |
| TM-F-010..012 | Task CRUD/status | Required fields, edit/delete, valid statuses, persisted refresh. |
| TM-F-020..023 | Project membership/assignment/board | Membership permission, assignees are members, correct columns and persistence. |
| TM-F-030..032 | Notifications/comments | Assignment event, due reminder/deduplication, comment access and persistence. |
| TM-F-040..042 | Search/report/export | Combined filters, accurate aggregates, authorized and correctly escaped CSV. |
| TM-NF-001/005 | Performance/concurrency | p95 API latency and 100 simulated users with profile/evidence. |
| TM-NF-002 | Availability | Report actual monitoring window and limitation; not fully provable in short test window. |
| TM-NF-003/004/006; PRJ-SR-001..006 | Security, accessibility, maintainability | Secrets/role checks/input tests/transport validation/keyboard and contrast/CI checks. |

# 4. Features Not to be Tested

- Offline-first behavior and native mobile application.
- Third-party email transport internals (email is future scope).
- Hosting provider control plane and infrastructure internals; application-facing deployment configuration is still checked.
- Browser engines outside the documented target set Chrome, Firefox and Edge.
- Availability target across a full month, unless long-running monitoring evidence is actually collected.

# 5. Test Strategy

## 5.1 Levels and Methods

| Level/type | Methods and focus | Tools (proposed) |
|---|---|---|
| Unit | Pure service logic, validation schemas, authorization helpers, UI components and utility functions. | Jest/Vitest; React Testing Library |
| API integration | HTTP status/body, middleware, services, repository persistence, duplicate/invalid references, database cleanup. | Jest/Vitest + Supertest; dedicated MongoDB test DB or MongoDB Memory Server if compatible |
| System / E2E | Register/login; project create/add member/create & assign task/status change/refresh/notification; search/report/export. | Playwright (preferred single baseline) or Cypress; choose one and standardize |
| Security | Password hash and no leakage, invalid/expired JWT, horizontal/role access, NoSQL/XSS and CSV injection, rate limiting, TLS config. | Automated API tests, code review, dependency/security scan, deployment config review |
| Performance | Run fixed request mix at normal load and 100 simulated concurrent users; capture p50/p95, error rate, throughput, environment. | k6 or JMeter |
| Usability/accessibility | Desktop/mobile viewport, keyboard flow, focus order, labels, errors, contrast, responsive layout. | Manual checklist + axe accessibility checks where available |
| Regression/CI | All unit/API tests, lint/build and E2E smoke suite on pull request and before release. | GitHub Actions; Jira for work/defects |

## 5.2 Entry Criteria

- SRS/SAD baseline reviewed by the team; each planned test maps to an SRS requirement ID.
- Build starts locally and target test command is documented; environment variables/sample env file prepared with no secrets.
- Dedicated test database and reset/seed procedure available; synthetic accounts and fixtures ready.
- CI workflow can run deterministic unit tests and API integration tests; E2E target environment selected.

## 5.3 Exit Criteria

- 100% of planned cases have an explicit result (Pass/Fail/Blocked/Not Run) and evidence; excluded cases have a reason.
- No open critical defects; all high severity defects fixed or have an explicitly approved disposition.
- All high-priority functional requirements meet acceptance criteria; security access-boundary tests pass.
- Lint/build and required CI checks pass on the release candidate; RTM updated and final test summary reviewed.
- Performance/availability claims are supported by dated measurements; any target not demonstrated is recorded as a limitation, not labeled Pass.

## 5.4 Test Data and Isolation

- Synthetic users for Admin, Lead and Member roles; at least two separate projects to test isolation.
- Tasks across To Do/In Progress/Done, priority values, upcoming/past due dates, multiple assignees, comments and read/unread notifications.
- Include duplicate email, invalid IDs, non-member assignee, malformed payload, XSS-looking text, NoSQL operator payload and CSV formula-like values as negative test data.
- Never use production credentials or real personal data. Reset DB between integration runs; automated tests should not rely on execution order.

# 6. Test Environment and Tools

| Area | Baseline setup |
|---|---|
| Runtime | Pin supported Node.js version in project config and CI; install dependencies from lockfile. |
| Frontend | Next.js/React development/production build; Chrome, Firefox and Edge current stable where available. |
| Backend | Node.js/Express API on isolated test config with safe test JWT secret; API base URL injected by environment. |
| Database | Dedicated MongoDB test database or ephemeral test server; Mongoose schema/index setup; deterministic seed/cleanup scripts. |

| Area | Baseline setup |
|---|---|
| Testing | Jest/Vitest + React Testing Library, Supertest, Playwright, k6/JMeter, optional axe, GitHub Actions and Jira. |
| Evidence | Test report/artifacts, CI run URL, screenshot/video on E2E failure, load-test output, sanitized logs and defect IDs. |

# 7. Test Cases

The following test catalogue is the initial planned baseline. Each Jira test/bug should retain the exact ID. Preconditions and expected results should be refined in automation scripts once the actual implementation endpoints and UI labels are frozen.

| Test ID | Req IDs | Objective | Preconditions | Steps (summary) | Expected result | Level |
|---|---|---|---|---|---|---|
| TC-AUTH-01 | TM-F-001 | Registration accepts valid fields | No user with email exists | POST valid name/email/password. | 201 created; one User record exists; passwordHash differs from plaintext; safe DTO has no passwordHash. | API + DB |
| TC-AUTH-01-N | TM-F-001 | Reject duplicate/invalid registration | Account exists for email | Register duplicate email; then malformed email/missing field. | 409 duplicate; 400 invalid payload; no extra record. | API |
| TC-AUTH-02 | TM-F-002 | Successful login and session | Registered test user | Login with correct email/password; call protected /me. | 200 and valid expiring session; protected response returns current safe profile. | API |
| TC-AUTH-02-N | TM-F-002, PRJ-SR-002 | Invalid/expired token rejected | Protected endpoint available | Call without token, malformed token, tampered token and expired token. | Each protected request returns 401; no protected data returned. | API security |
| TC-AUTH-03 | TM-F-003 | Profile/password update persists | Logged in user | Update permitted name; change password following documented flow; log in again. | Name persists; old password fails/new password succeeds; stored hash remains non-plaintext. | API + UI |
| TC-PROJ-01 | TM-F-020 | Create/edit project | Lead account active | Create project with valid name, update description. | 201 then updated details persisted; project appears only to authorized members. | API + DB |
| TC-PROJ-02 | TM-F-021, PRJ-SR-003 | Project membership rules | Lead, two members and outside user | Add registered member; attempt invalid user; attempt update by member/outside user. | Valid add works; invalid reference rejected; unauthorized mutation returns 403. | API security |
| TC-TASK-01 | TM-F-010 | Create task and validate required fields | Lead and project/member setup | Create task with title/due date/priority/project/assignee; omit required title in negative request. | Created task persists, status default documented; invalid request returns 400. | API + DB |
| TC-TASK-02 | TM-F-011 | Edit and delete task | Task exists in authorized project | Edit title/description; delete task and re-query. | Edits persist; deletion no longer appears; dependent comment/notification behavior matches policy. | API + DB |
| TC-TASK-03 | TM-F-012 | Task status validation | Task exists | Change to In Progress then Done; submit invalid status. | Valid states persist/update updatedAt; invalid value returns 400. | API |
| TC-BOARD-01 | TM-F-022 | Assign only project members | Project has one member and one outsider | Assign member; attempt outsider ID. | Member assignment accepted and notification created; outsider assignment rejected. | API + DB |
| TC-BOARD-02 | TM-F-023 | Kanban columns match persisted states | Tasks exist in all three states | Open board, change status, reload. | Each task appears once in correct column; after reload matches persisted status. | E2E |
| TC-NOTIF-01 | TM-F-030 | Assignment creates private notification | Task and assignee exist | Assign task, fetch notifications as assignee and unrelated user. | Assignee has relevant notification; unrelated user cannot access it. | API + DB |
| TC-NOTIF-02 | TM-F-031 | Due reminder rule/deduplication | Reminder rule configured; due soon and Done tasks seeded | Run reminder job twice with same inputs. | Eligible task recipient receives reminder; Done task excluded; duplicate policy prevents duplicate notification. | Service + DB |
| TC-COMM-01 | TM-F-032, PRJ-SR-005 | Comments persist and reject unsafe payloads | Member in project | Post normal comment; empty/oversized and XSS-looking text; request as outsider. | Valid comment persists with author/time; invalid rejected; text rendered safely; outsider denied. | API + UI |
| TC-SEARCH-01 | TM-F-040 | Search and filter combination | Diverse tasks seeded | Search keyword and combine status/priority/assignee filters. | Only authorized tasks satisfying all active filters are returned; empty result handled. | API + UI |
| TC-REPORT-01 | TM-F-041 | Dashboard counts and overdue total | Known fixed task set | Query summary; independently calculate expected counts. | Status counts/overdue values match seeded data and authorized project only. | API + DB |
| TC-REPORT-02 | TM-F-042, PRJ-SR-005 | CSV export correct and safe | Project contains commas, quotes, newlines and formula-like title | Export CSV; parse row count/header and inspect formula cells. | CSV fields escaped; expected rows only; unsafe spreadsheet formula prefixes neutralized according to chosen policy. | API |
| TC-SEC-01 | TM-NF-003, PRJ-SR-001 | Credential non-disclosure | Registered user and access to test DB/logs | Inspect DB, registration/login responses and logs. | Only bcrypt hashes stored; no password/token logged; passwordHash omitted from API DTO. | Code/DB/log review |
| TC-SEC-03 | PRJ-SR-003 | Role and horizontal access matrix | Admin, Lead, member, outsider; | For project, task, comment, export and notification endpoints, try allowed and forbidden read/write | Allowed actions succeed; forbidden return documented 403/404 consistently; no data | API security |

| Test ID | Req IDs | Objective | Preconditions | Steps (summary) | Expected result | Level |
|---|---|---|---|---|---|---|
| | | | two projects | combinations. | leaks. | |
| TC-SEC-04 | PRJ-SR-004 | Transport security in deployment | HTTPS staging available | Inspect certificate/protocol; call API over HTTP. | TLS 1.2+; HTTP redirected/rejected per deployment config. | Deployment |
| TC-SEC-05 | PRJ-SR-005 | Injection and input validation | Test environment isolated | Submit NoSQL operator payload, excessive strings, malformed ID, HTML/script text and invalid date. | Invalid values rejected or safely handled; no unauthorized bypass, injection, script execution or stack leakage. | API + UI |
| TC-SEC-06 | PRJ-SR-006 | Rate limiting and safe server error | Test environment isolated | Repeated authentication attempts; trigger controlled unexpected error in non-production test. | Rate limit applies (429 or configured policy); response omits stack/secret; logs omit credential/token. | API |
| TC-UX-01 | TM-NF-004 | Responsive and accessible core flow | Supported browsers available | Use registration/login/project/board/task flows at desktop/mobile widths, keyboard only. | No critical overflow; labels and focus visible; modal/forms usable; readable contrast and status errors. | Manual + accessibility scan |
| TC-PERF-01 | TM-NF-001 | Normal-load API latency | Stable staging and defined workload | Run k6/JMeter for agreed duration/request mix; collect percentile metrics. | At least 95% of sampled API requests complete ≤2s (p95 ≤2s); report errors and environment. | Load test |
| TC-PERF-02 | TM-NF-005 | 100 simulated concurrent users | Stable staging and dataset | Run 100 virtual users with representative reads/writes; capture p95/error rate/throughput. | No unacceptable degradation under agreed workload; record measurable thresholds and evidence before pass. | Load test |
| OPS-MON-01 | TM-NF-002 | Availability observation report | Monitoring available | Collect environment uptime for stated window and record maintenance/outage events. | Report measured value/window and limitations; 99% monthly target remains unverified until adequate window. | Operations |
| TC-CI-01 | TM-NF-006 | CI pipeline checks | PR raised | Open/update feature PR. | Install/lint/test/build jobs pass; test artifact/status retained; merge follows team review convention. | GitHub Actions |
| TC-E2E-01 | TM-F-001/002, TM-F-020..023, TM-F-030 | Critical user journey | Clean test environment | Register/login; create project; add member; create/assign task; update status; reload; view notification. | Full journey completes; data, board state and notification are persisted/correct. | Playwright |

## 8. Execution Schedule and Milestones

Deadline: 30 October 2026. Suggested compressed schedule from the current date (15 October 2026); adjust in Jira based on team availability and implementation maturity.

| Dates | Milestone | Exit evidence | Suggested owner |
|---|---|---|---|
| 15-Oct | Baseline and scope freeze | Team review of SRS/SAD/STP; unresolved decisions logged in Jira. | All; coordinator rotates |
| 16–18 Oct | Scaffold and CI foundation | Build/lint/test scripts; env example; test DB setup. | Member 1 + Member 4 review |
| 18–23 Oct | Feature implementation in parallel | PRs linked to Jira stories; unit/API tests per feature. | All feature owners; swap reviewers |
| 22–25 Oct | Integration and regression | API/UI integrated; P0/P1 tests; data/access checks. | Member 2 + Member 3 coordinate |
| 25–27 Oct | System, security and usability testing | E2E proof, negative RBAC, UI checks, defect fixes. | All; rotate test executor |
| 27–28 Oct | Performance and final regression | k6/JMeter report; repeat critical suite; CI all green. | Performance owner + peer reviewer |
| 28–29 Oct | Documentation and release evidence | RTM/test summary, SAD UML render, demo data, deployment notes. | Documentation owner + team |
| 30-Oct | Submission and demo | Tagged candidate, final report/presentation, sign-off or limitations. | All team members |

## 9. Team Roles and Responsibilities

Roles are initial coordination assignments only and may rotate. Every member should contribute code/review/testing/documentation. Avoid having the author be the only reviewer of their own change.

| Member | Initial focus | Shared duties |
|---|---|---|
| Akash Kiran Vernekar (PES1UG24AM338) | API/backend integration and technical coordination | Review another member's PR; API tests; demo integration. |
| Vismay H S (PES1UG24AM325) | Test strategy, test data and CI test evidence | Implement features or fixes; execute assigned tests; record Jira defects. |
| Basawaraj Panchal (PES1UG24AM350) | Product scope, Jira backlog and acceptance criteria | Implement project/task feature slice; UAT and report accuracy. |

| Member | Initial focus | Shared duties |
|---|---|---|
| Trisha G Shetty (PES1UG24AM308) | Next.js UI and documentation coordination | Implement UI feature slice; accessibility/E2E tests; maintain SRS/SAD/STP links. |

## 10. Defect Management

- Create Jira Bug with reproducible title/steps, expected vs actual, environment/build/commit, severity, screenshots/logs and linked requirement/test ID. Never attach secrets or personal data.
- Suggested severity: Critical (data breach/data loss/system unusable); High (major required feature or authorization failure); Medium (workaround exists or isolated feature defect); Low (minor UX/cosmetic).
- Triage daily during integration. Fix on branch; add regression test where possible; review and rerun failed test plus relevant regression suite before closing.
- A failed CI run is not automatically a product defect; record flaky/environment causes distinctly and provide rerun evidence.

## 11. Risks and Mitigation

| Risk | Mitigation |
|---|---|
| Unstable/late implementation | Prioritize high-priority vertical slices and critical journeys; define a clear minimal release and defer future scope. |
| Schema drift or dangling references | Mongoose validation; explicit deletion policy; database integrity fixtures; schema-impact note in PR. |
| Authorization gaps | Negative access tests for each entity/endpoint; backend middleware and service-level membership checks. |
| Flaky E2E test | Use stable test selectors, seeded isolated DB and repeatable API setup/cleanup; record failure artifacts. |
| Performance infrastructure too limited | Use local/staging controlled run and report scope honestly; never infer monthly uptime from short test. |
| Conflicts/integration debt | Short-lived feature branches, early PRs, one owner per module with cross-review and daily integration. |

## 12. Suspension and Resumption Criteria

Suspend test execution if the test environment is unavailable for more than four hours, a build blocks more than 30% of planned tests, test data corruption makes results unreliable, or a suspected active security/data-loss issue could spread. Resume when blocking defects are fixed or isolated, the environment is stable, and the seed/reset procedure produces known state.

## 13. Traceability and Result Recording

SRS §8 is the canonical RTM; the following summarizes each SRS group and required evidence. Results start as Planned. After execution record status, date, executor, environment/build commit, evidence link and defect ID.

| SRS requirements | Test cases / evidence | Planned status |
|---|---|---|
| TM-F-001..003 | TC-AUTH-01/02/03; TC-SEC-01/02/06 | Planned |
| TM-F-010..012 | TC-TASK-01/02/03 | Planned |
| TM-F-020..023 | TC-PROJ-01/02; TC-BOARD-01/02; TC-E2E-01 | Planned |
| TM-F-030..032 | TC-NOTIF-01/02; TC-COMM-01 | Planned |
| TM-F-040..042 | TC-SEARCH-01; TC-REPORT-01/02 | Planned |
| TM-NF-001 / TM-NF-005 | TC-PERF-01 / TC-PERF-02 | Planned |
| TM-NF-002 | OPS-MON-01 | Planned / long-window evidence required |
| TM-NF-003; PRJ-SR-001/002 | TC-SEC-01 / TC-AUTH-02-N | Planned |
| PRJ-SR-003 | TC-PROJ-02; TC-SEC-03 | Planned |
| PRJ-SR-004/005/006 | TC-SEC-04/05/06 | Planned |
| TM-NF-004 / TM-NF-006 | TC-UX-01 / TC-CI-01 | Planned |

## 14. Metrics and Reporting

- Planned/executed/passed/failed/blocked test counts and execution percentage.
- Pass rate; open defects by severity; defect aging; escaped regressions.
- Requirement coverage: in-scope SRS requirement IDs with at least one linked test and execution evidence.
- For load runs: duration, virtual users, requests/sec, p50/p95 latency, error rate, CPU/memory if available, commit and environment.
- For reliability: observation window and measured uptime; clearly identify when monthly target cannot yet be established.
- Daily brief during final sprint; final Test Summary Report with release recommendation and known limitations.

## 15. Test Deliverables and Approval

- This STP; detailed test scripts; synthetic seed data; automated unit/API/E2E tests; sanitized CI artifacts; defect records; accessibility checklist; performance report; RTM updates; final test summary.

| Role | Name | Approval / date |
|---|---|---|
| Developer / integrator | Akash Kiran Vernekar (PES1UG24AM338) | Pending |
| QA coordination | Vismay H S (PES1UG24AM325) | Pending |
| Product/backlog | Basawaraj Panchal (PES1UG24AM350) | Pending |
| UI/documentation | Trisha G Shetty (PES1UG24AM308) | Pending |
| Course instructor | TBD | Pending |

## Appendix A. Run Result Template

| Field | Value to record |
|---|---|
| Test ID / requirement | e.g., TC-SEC-03 / PRJ-SR-003 |
| Status | Pass / Fail / Blocked / Not Run |
| Date, executor | Actual date and teammate |
| Build/environment | commit SHA, Node version, target URL/environment label |
| Evidence | CI run URL, screenshot, sanitized log, test output |
| Defect | Jira Bug key if failed; link regression test |
| Notes | Deviation, assumptions, limitations |